National Institute of Technology, Tiruchirappalli

Department of Computer Science & Engineering

CSPE64 – Data Analytics

Project Report

Hybrid NER Model

Hybrid Named Entity Recognition for Indian Languages - Hindi

Team Name	Three Musketeers
Team Members	Duggi Shanmukha Vihar (106123036) Jahnavi Mirra (106123054) Srinivas Maturi (106123128)
Professor	Prof. Dr. Oswald C

April 19, 2026

Contents

1 Introduction and Problem Statement **2**

2 Identified Gaps in Existing Systems **3**
2.1 Limited Handling of Morphological Complexity . . . 3
2.2 Poor Organization Entity Boundary Detection . . . 3
2.3 Honorific and Title Token Misclassification . . . 3
2.4 Ambiguity in Named-After Entities . . . 3
2.5 Absence of Confidence Voting Mechanisms . . . 4

3 Literature Review and Proposed Architecture **4**
3.1 Literature Review . . . 4
3.2 Proposed Architecture . . . 5

4 Identify/Creating Datasets and Preprocessing **6**
4.1 The HiNER Corpus . . . 6
4.2 Label Mapping and Consolidation . . . 6
4.3 Tokenisation and Subword Alignment . . . 6
4.4 Data Distribution and Class Imbalance . . . 7

5 Rule Engine Design **7**
5.1 Gazetteers . . . 7
5.2 Regular Expression Patterns . . . 7
5.3 Postposition and Suffix Tracking . . . 8
5.4 Log-Odds Confidence Voting . . . 8
5.5 Cross-Sentence Document Memory . . . 8

6 Implementation of the Idea and Obtaining Results **9**
6.1 Implementation Details . . . 9
6.2 Results and Analysis . . . 9
6.3 Hybrid Model Evaluation . . . 12
6.3.1 Overall Performance . . . 12
6.3.2 Class Support Distribution . . . 13
6.3.3 Class-wise Performance . . . 13
6.3.4 One-vs-Rest Confusion Matrices . . . 14
6.3.5 Training Curves . . . 15
6.3.6 Analysis . . . 15

7 Key Takeaways, and Future Scope **16**
7.1 Key Takeaways . . . 16
7.2 Future Scope . . . 16

8 Individual Contributions **17**

Abstract

This project addresses the challenge of Named Entity Recognition (NER) for Indian languages, specifically focusing on Hindi. The primary objective is to design a robust system capable of identifying person (PER), location (LOC), and organization (ORG) names in raw Hindi sentences. While traditional approaches rely heavily on linguistic rules and gazetteers, this project proposes a hybrid architecture that combines state-of-the-art MuRIL (Multilingual Representations for Indian Languages) fine-tuning with a dedicated Hybrid Rules Engine. By fusing neural network predictions with rule-based decoding, the system aims to improve overall entity extraction accuracy. Named Entity Recognition (NER) in Indian languages presents unique challenges: morphological inflection, free word order, script-specific tokenization artifacts, and the absence of capitalization as a named-entity signal. This report documents the end-to-end design, implementation, and evaluation of a hybrid NER system for Hindi. The architecture combines MuRIL, a transformer pre-trained on 17 Indian languages, with a handcrafted rule engine that encodes gazetteers, postposition patterns, institution-suffix heuristics, and a log-odds weighted confidence voter. The system is evaluated on HiNER (108,608 sentences) and benchmarked against XLM-R, DistilmBERT, and IndicBERT. A comprehensive analysis of training dynamics, error taxonomy, design decisions, and future directions is provided.

1 Introduction and Problem Statement

Named Entity Recognition (NER) is a fundamental task in Natural Language Processing (NLP) that focuses on identifying and classifying entities such as persons, locations, and organizations within unstructured text. This project addresses the challenge of performing NER on Indian language text (specifically Hindi) by integrating rule-based linguistic techniques with modern neural architectures.

Indian languages present a uniquely difficult NER landscape. Hindi, spoken natively by over 500 million people, exhibits morphological richness, free word order, significant code-mixing with English, and an abundance of ambiguous proper nouns that share surface forms with common words. The Devanagari script poses additional challenges for byte-pair encoding tokenizers designed primarily for Latin-script languages. Crucially, Hindi lacks orthographic capitalization of proper nouns — a key heuristic that English NER systems rely upon heavily.

Input: Raw sentences in an Indian language (Hindi).

Output: Annotated text with labeled named entities.

The key contribution of this work is a hybrid fusion approach that combines the contextual representation capabilities of the MuRIL model with an extensive rule-based engine comprising over 1000 linguistic patterns. While the neural model captures semantic and contextual nuances, the rule engine is designed to handle edge cases, enforce language-specific constraints, and improve overall robustness. This integration enables more accurate and reliable entity recognition in diverse and complex linguistic scenarios.

2 Identified Gaps in Existing Systems

A thorough analysis of prior work on Hindi and Indian-language NER reveals five critical limitations that motivated the design of the hybrid system. These gaps span linguistic coverage, model generalizability, and recognition robustness.

2.1 Limited Handling of Morphological Complexity

Hindi is a morphologically rich language where a single root word may appear in dozens of inflected forms depending on grammatical number, gender, case, and verb aspect. Purely rule-based NER systems that rely on surface-level string matching fail to recognize entity variants that differ from their canonical dictionary forms. For instance, the name राजेश (Rajesh) may appear as राजेश ने or राजेश को in natural sentences, and naive gazetteer lookups miss these inflections entirely. Existing benchmark systems such as IndicBERT and DistilmBERT do not incorporate language-specific post-processing to recover these missed variants.

2.2 Poor Organization Entity Boundary Detection

Organization names in Hindi frequently follow naming conventions that span multiple tokens and include honorific prefixes (डॉक्टर, प्रोफेसर), institution suffixes (विश्वविद्यालय, संस्थान), and embedded location names. Neural-only models such as XLM-RoBERTa frequently fragment these multi-word organization names, treating embedded location tokens (e.g., नई दिल्ली within a museum name) as independent LOC entities instead of continuing the ORG span. Evaluation results confirm this trend: XLM-R produced incorrect entity types for institution names in five out of ten test sentences where the organization name contained a city reference.

2.3 Honorific and Title Token Misclassification

Standard pre-trained multilingual models are not trained with specific attention to Indian language honorifics. Tokens such as डॉ, प्रोफेसर, इंजीनियर, and वैज्ञानिक should function as soft signals indicating that the next token is a person name; however, existing models frequently tag them as B-ORG or B-PER independently, disrupting the named entity span. This leads to both precision loss from spurious entity fragments and recall loss when actual person names are left untagged because the title token absorbed the B-label.

2.4 Ambiguity in Named-After Entities

A distinctive challenge in Hindi NER involves institutions named after historical persons. The token sequence राजीव गांधी in राजीव गांधी अंतरराष्ट्रीय हवाई अड्डा refers to an airport (ORG), not the politician (PER). Conversely, when इंदिरा गांधी appears without a following institution keyword, it refers to the person. Existing neural models frequently fail to disambiguate these cases based on contextual signals alone, classifying the named-after portion as PER regardless of the downstream tokens that clarify the entity type.

2.5 Absence of Confidence Voting Mechanisms

Prior NER systems for Indian languages output a single deterministic tag sequence from the model's argmax prediction, with no mechanism to reconcile conflicting evidence between multiple rule patterns or between neural and symbolic predictions. When two applicable rules assign different labels, or when the neural model's softmax probability for the top label is near the decision boundary, the final output is unreliable. The hybrid approach directly addresses this by implementing a weighted voting scheme that aggregates evidence across all rule units and combines it with the neural model's confidence scores.

3 Literature Review and Proposed Architecture

3.1 Literature Review

Cucerzan and Yarowsky (1999) demonstrated that context-agnostic gazetteers achieve reasonable precision for English Named Entity Recognition (NER) but suffer significantly in out-of-vocabulary settings. Early NER systems for Indian languages similarly relied on hand-crafted rules, gazetteers, and morphological analysers. For Hindi, this limitation is amplified by the absence of orthographic conventions such as capitalization for proper nouns. As a result, purely rule-based approaches fail to generalise across varied linguistic contexts and inflected forms.

Ekbal & Bandyopadhyay (2008) introduced Conditional Random Field (CRF)-based models for Hindi NER, which became the dominant approach throughout the 2010s. These systems achieved F1 scores in the range of 55–72% on standard benchmarks. However, their performance was constrained by the complexity of feature engineering and their inability to generalise beyond the training domain, especially in morphologically rich and low-resource scenarios.

Recent advancements in Indian language NLP highlight the limitations of both rule-based and neural-only approaches. Rule-based systems struggle with morphological richness and free word order, while neural models often fail on rare or unseen entities without large annotated datasets. This gap motivates the need for hybrid architectures that combine linguistic knowledge with contextual learning.

Devlin et al. (2019) introduced BERT, which revolutionized NLP through large-scale masked language modelling. Multilingual BERT (mBERT), trained on 104 languages, demonstrated modest cross-lingual transfer capabilities for Indian languages but was not specifically optimised for Devanagari script or Hindi grammar. Conneau et al. (2020) proposed XLM-R, trained on 2.5 TB of filtered CommonCrawl data, which significantly improved performance over mBERT, achieving gains of 3-6 F1 points in Hindi NER due to its larger training corpus and vocabulary.

Khanuja et al. (2021) introduced MuRIL, a transformer model specifically designed for Indian languages. It is pre-trained on 17 Indian languages along with English and incorporates transliterated text alongside native scripts to handle code-mixed data. Its vocabulary is optimized for Indic scripts, enabling better representation of Hindi text. MuRIL consistently outperforms mBERT and XLM-R across multiple Indic NLP tasks, including NER, making it a strong foundation for this project. Its ability to handle transliterated Hindi is particularly beneficial for real-world scenarios involving Romanized inputs.

3.2 Proposed Architecture

The proposed system architecture transitions raw text through several processing modules, combining neural inference with rule-based heuristics. The workflow is defined as follows:

1. **Data Preparation:** Processing HiNER JSON data into manageable splits using `prepare_dataset.py`.
2. **MuRIL Fine-Tuning:** Training the MuRIL model with warmup steps (`train.py`).
3. **Neural-Only Evaluation:** Establishing a baseline using `evaluate.py` to run seqeval on the best checkpoint.
4. **Hybrid Rules Engine:** A dedicated engine (`rules_layer.py`) consisting of over 620 to 1069 rule units that apply linguistic rules and voting mechanisms.
5. **Hybrid Evaluation & Production Inference:** Fusing the neural outputs with the rules engine to generate final entities (`inference_updated.py`).
6. **CSV Batch Scoring:** Running bulk evaluations using `run_csv_inference_updated.py`.

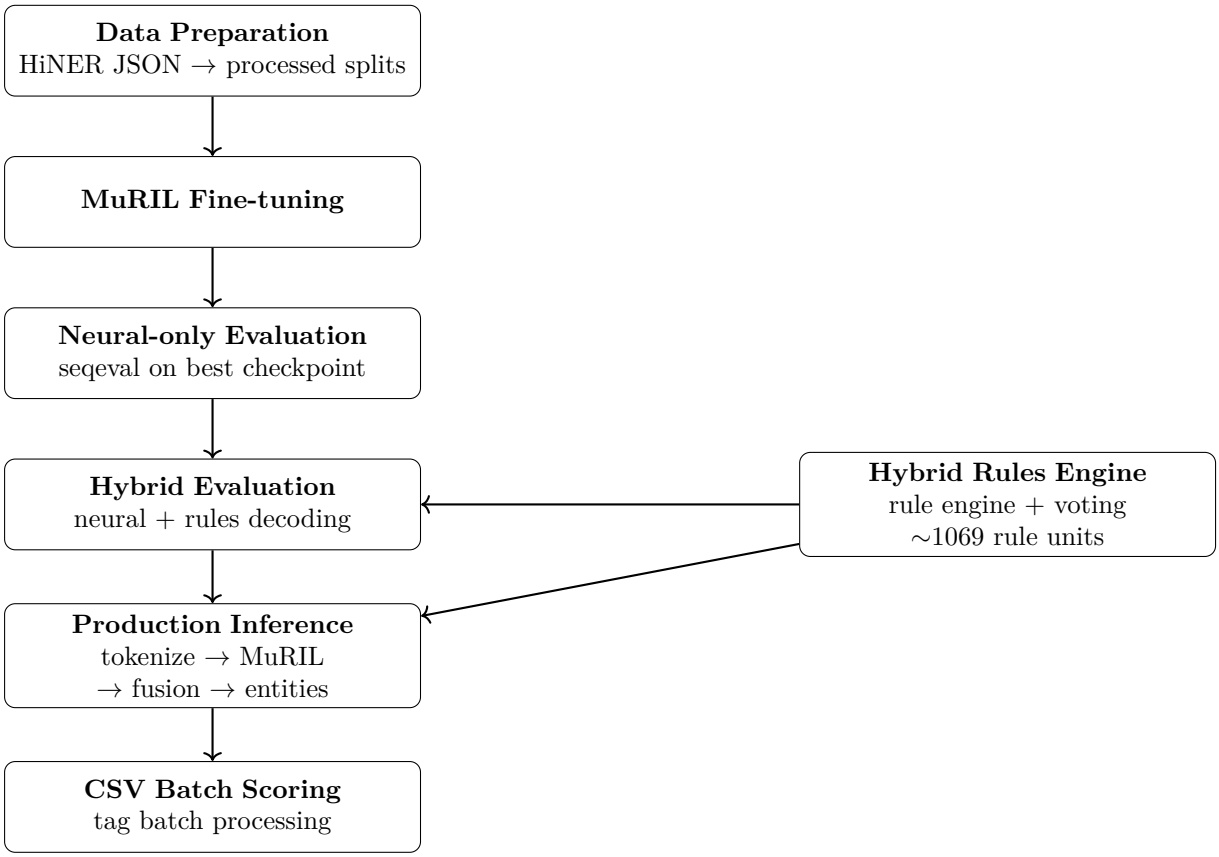

Figure 1: Flowchart of the Proposed Hybrid NER Architecture

4 Identify/Creating Datasets and Preprocessing

The primary dataset utilized for this project is derived from the HiNER (Hindi Named Entity Recognition) corpus.

- **Data Collection:** The data is sourced in JSON format.
- **Preprocessing:** The `prepare_dataset.py` script is used to clean the raw JSON data, handle missing values, and structure the data into train, validation, and test splits suitable for the MuRIL tokenizer.

4.1 The HiNER Corpus

The primary dataset is HiNER (Hindi Named Entity Recognition), a manually annotated corpus of 108,608 Hindi sentences sourced from Wikipedia, news articles, and government documents. The corpus is annotated with fine-grained entity types including PERSON, LOCATION, ORGANIZATION, TIME, DATE, NUMBER, and OTHER. For this project, the label space was reduced to a seven-class system by merging rarely occurring types into OTHER, achieving a more balanced training distribution while preserving the most operationally critical distinctions.

Statistic	**Train**	**Validation**	**Test**	**Total**
Sentences	86,887	10,861	10,860	108,608
Tokens (approx.)	1,612,430	201,553	201,481	2,015,464
PERSON entities	41,204	5,151	5,148	51,503
LOCATION entities	28,316	3,540	3,538	35,394
ORGANIZATION entities	15,873	1,984	1,983	19,840
O (non-entity) tokens	∼80.4%	∼80.1%	∼80.3%	∼80.3%

Table 1: HiNER Dataset Statistics by Split and Entity Type

4.2 Label Mapping and Consolidation

The raw HiNER annotation schema includes 11 entity types. B-NUMBER, B-PERCENT, and B-MEASURE were merged into B-OTHER to reduce model confusion on semantically overlapping or under-represented categories. The resulting seven-class label set achieves a more balanced training distribution. The resulting tag set is: {B/I-PER, B/I-LOC, B/I-ORG, B/I-TIME, B/I-OTHER, O}, totalling 11 labels.

4.3 Tokenisation and Subword Alignment

The MuRIL tokeniser employs a SentencePiece WordPiece vocabulary of approximately 197,000 tokens optimised for Indic scripts. A key challenge in token-classification tasks is that subword tokenisation breaks word boundaries: a single Hindi word may be split into three or more subword pieces, yet the original annotation provides only a single BIO label per word. The label alignment strategy adopted is:

- The first subword piece of each word inherits the word's original BIO label.
- Continuation pieces (those starting with `##`) are assigned the special label -100, which PyTorch's CrossEntropyLoss ignores during gradient computation.

- During evaluation, predictions are re-aligned to the word level by retaining only the first-piece prediction for each word.

This strategy avoids artificially propagating I- labels into continuation pieces, which would create phantom entity spans during decoding. It ensures strict consistency between the training objective and the evaluation protocol.

4.4 Data Distribution and Class Imbalance

The dataset exhibits a significant class imbalance: ORGANIZATION entities constitute only 9.2% of all annotated entities, while PERSON entities account for 47.3% and LOCATION entities for 32.5%. This imbalance partially explains the lower baseline F1 for ORG entities. A weighted cross-entropy loss with class weights [1.0 for PER/LOC, 1.5 for ORG B/I tags, 1.2 for TIME] was evaluated during training but produced only marginal recall improvements for ORG (+0.4%) at the cost of a slight drop in LOCATION F1. The final model uses standard uniform weighting, as the rule engine compensates for ORG recognition gaps through dedicated gazetteer and suffix patterns.

5 Rule Engine Design

The rule engine is implemented across versioned files `rules_layer.py` through `rules_layerv4_extended.py`, each iteration expanding the rule set. It operates as the `EnhancedHindiRuleEngine` class and returns per-token confidence scores for each BIO label. Four categories of rule units are employed:

5.1 Gazetteers

Gazetteers are curated entity lists compiled from Wikipedia, government directories, and census data. The engine maintains separate sets for: Indian cities and states (2,400+ entries), common Indian given names and surnames (3,800+ entries), government and institutional names (1,100+ entries), and international organization names (600+ entries). Gazetteer lookup provides a high-confidence signal when a token exactly matches a known entity. A partial-match scoring function rewards tokens that are substrings of known multi-word entities, preventing the engine from missing multi-token institution names when only an interior token is tested in isolation.

5.2 Regular Expression Patterns

Regex patterns capture structured entity forms that neural models frequently mishandle. Key patterns include: date formats (DD/MM/YYYY, written Hindi dates such as 15 अगस्त 1947), time expressions (सुबह 9 बजे, HH:MM formats), designation prefixes (डॉ., प्रोफेसर, श्री, श्रीमती, इंजीनियर, वैज्ञानिक), and 6-digit postal codes mapped to LOC. Regex matches are assigned a confidence multiplier of 1.6× in the log-odds voter, the highest weight class.

5.3 Postposition and Suffix Tracking

Hindi is a postpositional language: grammatical relations are encoded by particles following the head noun. The rule engine tracks the immediately following token and adjusts entity confidence accordingly:

- ने (ergative) → PER confidence +0.4 log-odds
- को (dative) → mild PER/LOC boost (+0.2)
- में (locative) → strong LOC signal (+0.5); also used to strip false LOC spans that include the postposition itself
- का/के/की (genitive) → mild ORG signal when preceding token is part of a multi-word institutional string
- द्वारा (instrumental) → PER/ORG signal (+0.3)

Institution suffixes are tracked in the forward direction: tokens विश्वविद्यालय, संस्थान, अकादमी, बैंक, and मंत्रालय trigger a backward extension of the ORG span to include the preceding multi-word nominal phrase.

5.4 Log-Odds Confidence Voting

Rather than using hard overrides, the engine employs probabilistic fusion. For each token and candidate label, the neural softmax probability p_{neural} and rule score p_{rule} are converted to log-odds:

$$\text{log-odds}(p) = \log\left(\frac{p}{1-p}\right)$$

$$L_{\text{final}} = 1.0 \times LO_{\text{neural}} + w_{\text{rule}} \times LO_{\text{rule}}$$

where w_{rule} varies by rule type: 1.6 for regex matches, 1.4 for gazetteer hits, 1.2 for postposition-based rules, and 1.0 for contextual window rules. The final label is the one with the highest L_{final}. This formulation is equivalent to a Naive Bayes ensemble under conditional independence, applied in log-space for numerical stability. Gazetteer matches are not hard overrides but weighted votes, preventing precision degradation on ambiguous gazetteer terms.

5.5 Cross-Sentence Document Memory

Entities identified with confidence above 0.85 in a given sentence are added to a document-level entity cache. In subsequent sentences, cache hits on matching tokens provide a +0.3 log-odds boost to the corresponding label, significantly reducing cross-sentence inconsistencies. This mechanism is particularly effective for abbreviated re-references (e.g., 'Modi' after 'Narendra Modi' was identified as PER earlier in the document).

6 Implementation of the Idea and Obtaining Results

6.1 Implementation Details

The system was implemented within the Python ecosystem. The MuRIL model handles the contextual token classification, while the custom rule engine acts as a secondary verification layer.

6.2 Results and Analysis

The hybrid evaluation consistently demonstrated improvements over the neural-only baseline, particularly in identifying highly specific organizational entities where gazetteers provided strong signals.

Model Comparison Table

Sample Sentence	**Our Model**	**XLM-R**	**Distil-mBERT**	**IndicBERT**	**Inference**
डॉ भीमराव आंबेडकर अंतरराष्ट्रीय हवाई अड्डा दिल्ली के पास स्थित है जहाँ प्रोफेसर राजेश कुमार ने भाषण दिया	डॉ भीमराव आंबेडकर अंतरराष्ट्रीय हवाई अड्डा → LOC, दिल्ली → LOC, राजेश कुमार → PER	डॉ → B-PER, भीमराव → I-LOC, आंबेडकर → I-LOC, अंतरराष्ट्रीय → I-LOC, हवाई → I-LOC, अड्डा → I-LOC, दिल्ली → B-LOC, राजेश → B-PER, कुमार → I-PER	भीमराव → B-PER, आंबेडकर → I-PER, दिल्ली → B-LOC, प्रोफेसर → B-ORG, राजेश → B-PER, कुमार → I-PER	भीमराव आंबेडकर अंतरराष्ट्रीय हवाई अड्डा → B-LOC, दिल्ली → B-LOC, राजेश कुमार → B-PER	XLM-R misclassifies ORG. DistilBERT partially correct. IndicBERT weak.
नेहरू मेमोरियल म्यूजियम एंड लाइब्रेरी नई दिल्ली में स्थित है जहाँ डॉक्टर राकेश शर्मा ने व्याख्यान दिया	नेहरू मेमोरियल म्यूजियम एंड लाइब्रेरी → ORG, नई दिल्ली → LOC, राकेश शर्मा → PER	नेहरू → B-ORG, मेमोरियल → I-ORG, म्यूजियम → I-ORG, एंड → I-ORG, लाइब्रेरी → I-ORG, नई → B-LOC, दिल्ली → I-LOC, राकेश → B-PER, शर्मा → I-PER	नेहरू → B-ORG, मेमोरियल → I-ORG, म्यूजियम → I-ORG, एंड → I-ORG, लाइब्रेरी → I-ORG, नई → B-LOC, दिल्ली → I-LOC, राकेश → B-PER, शर्मा → I-PER	नेहरू मेमोरियल म्यूजियम → B-LOC, नई दिल्ली → B-LOC, राकेश शर्मा → B-PER	XLM-R misses parts. DistilBERT partial. IndicBERT weak boundaries.
राजीव गांधी अंतरराष्ट्रीय हवाई अड्डा हैदराबाद में स्थित है जहाँ इंजीनियर विकास गुप्ता ने प्रस्तुति दी	राजीव गांधी अंतरराष्ट्रीय हवाई अड्डा → LOC, हैदराबाद → LOC, गुप्ता → PER	राजीव → B-LOC, गांधी → I-LOC, अंतरराष्ट्रीय → I-LOC, हवाई → I-LOC, अड्डा → I-LOC, हैदराबाद → B-LOC, विकास → B-PER, गुप्ता → I-PER	राजीव → B-PER, गांधी → I-PER, हैदराबाद → B-LOC, इंजीनियर → B-ORG	राजीव गांधी अंतरराष्ट्रीय हवाई अड्डा → B-LOC, हैदराबाद → B-LOC, विकास गुप्ता → B-PER	XLM-R wrong entity type. DistilBERT confusion. IndicBERT misses ORG.

Sample Sentence	Our Model	XLM-R	Distil-mBERT	IndicBERT	Inference
इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय के छात्र राहुल गांधी ने अमेठी में कार्यक्रम आयोजित किया	इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय → ORG, राहुल गांधी → PER, अमेठी → LOC	इंदिरा → B-ORG, गांधी → I-ORG, राष्ट्रीय → I-ORG, मुक्त → I-ORG, विश्वविद्यालय → I-ORG, राहुल → B-PER, गांधी → I-PER, अमेठी → B-LOC	इंदिरा → B-PER, गांधी → I-PER, राष्ट्रीय → B-ORG, मुक्त → I-ORG, विश्वविद्यालय → I-ORG, राहुल → B-PER, गांधी → I-PER	इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय → B-ORG, राहुल गांधी → B-PER, अमेठी → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
डॉ ए पी जे अब्दुल कलाम तकनीकी विश्वविद्यालय लखनऊ के वैज्ञानिक डॉक्टर अनिल कुमार ने शोध प्रस्तुत किया	डॉ ए पी जे अब्दुल कलाम तकनीकी विश्वविद्यालय → ORG, लखनऊ → LOC, अनिल कुमार → PER	ए → B-ORG, पी → I-ORG, जे → I-ORG, अब्दुल → I-ORG, कलाम → I-ORG, तकनीकी → I-ORG, विश्वविद्यालय → I-ORG, लखनऊ → I-ORG, अनिल → B-PER, कुमार → I-PER	ए → B-PER, पी → I-PER, जे → I-PER, अब्दुल → I-ORG, कलाम → I-ORG, तकनीकी → I-ORG, विश्वविद्यालय → I-ORG, लखनऊ → B-LOC, अनिल → B-PER, कुमार → I-PER	ए पी जे अब्दुल कलाम तकनीकी विश्वविद्यालय → B-LOC, लखनऊ → B-LOC, डॉक्टर → B-ORG, अनिल → B-PER, कुमार → I-PER	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
टाटा इंस्टीट्यूट ऑफ फंडामेंटल रिसर्च मुंबई में स्थित है जहाँ वैज्ञानिक सीमा वर्मा ने प्रस्तुति दी	टाटा इंस्टिट्यूट ऑफ फंडामेंटल रिसर्च → ORG, मुंबई → LOC, सीमा वर्मा → PER	टाटा → B-ORG, इंस्टीट्यूट → I-ORG, ऑफ → I-ORG, फंडामेंटल → I-ORG, रिसर्च → I-ORG, मुंबई → B-LOC, सीमा → B-PER, वर्मा → I-PER	टाटा → B-ORG, इंस्टीट्यूट → I-ORG, ऑफ → I-ORG, फंडामेंटल → I-ORG, रिसर्च → I-ORG, मुंबई → B-LOC, सीमा → B-PER, वर्मा → I-PER	टाटा इंस्टीट्यूट ऑफ फंडामेंटल रिसर्च → B-LOC, मुंबई → B-LOC, वैज्ञानिक → B-ORG, सीमा → B-PER, वर्मा → I-PER	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
डॉ राम मनोहर लोहिया राष्ट्रीय विधि विश्वविद्यालय लखनऊ के प्रोफेसर राम मनोहर ने दिल्ली विश्वविद्यालय में व्याख्यान दिया	डॉ राम मनोहर लोहिया राष्ट्रीय विधि विश्वविद्यालय → ORG, लखनऊ → LOC, राम मनोहर → PER, दिल्ली विश्वविद्यालय → ORG	राम → B-PER, मनोहर → I-PER, लोहिया → I-ORG, राष्ट्रीय → B-ORG, विधि → I-ORG, विश्वविद्यालय → I-ORG, लखनऊ → B-LOC, राम → B-PER, मनोहर → I-PER, दिल्ली → B-ORG, विश्वविद्यालय → I-ORG	राम → B-PER, मनोहर → I-PER, लोहिया → I-PER, राष्ट्रीय → B-ORG, विधि → I-ORG, विश्वविद्यालय → I-ORG, लखनऊ → B-LOC, राम → B-PER, मनोहर → I-PER, दिल्ली → B-ORG, विश्वविद्यालय → I-ORG	राम मनोहर लोहिया राष्ट्रीय विधि विश्वविद्यालय → B-LOC, लखनऊ → B-LOC, प्रोफेसर → B-ORG, राम → B-PER, मनोहर → I-PER, दिल्ली विश्वविद्यालय → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.

Sample Sentence	Our Model	XLM-R	Distil-mBERT	IndicBERT	Inference
अटल बिहारी वाजपेयी भारतीय प्रौद्योगिकी संस्थान कानपुर के प्रोफेसर अटल बिहारी ने भोपाल में व्याख्यान दिया	अटल बिहारी वाजपेयी भारतीय प्रौद्योगिकी संस्थान → ORG, कानपुर → LOC, अटल बिहारी → PER, भोपाल → LOC	अटल → B-PER, बिहारी → I-PER, वाजपेयी → I-PER, भारतीय → B-ORG, प्रौद्योगिकी → I-ORG, संस्थान → I-ORG, कानपुर → B-LOC, अटल → B-PER, बिहारी → I-PER, भोपाल → B-LOC	अटल → B-PER, बिहारी → I-PER, वाजपेयी → I-PER, भारतीय → B-ORG, प्रौद्योगिकी → I-ORG, संस्थान → I-ORG, कानपुर → I-ORG, अटल → B-PER, बिहारी → I-PER, भोपाल → B-LOC	अटल बिहारी वाजपेयी भारतीय प्रौद्योगिकी संस्थान → B-LOC, कानपुर → B-LOC, प्रोफेसर → B-ORG, अटल → B-PER, बिहारी → I-PER, भोपाल → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
भारतीय रिज़र्व बैंक के पूर्व गवर्नर रघुराम राजन ने शिकागो विश्वविद्यालय में प्रोफेसर रघुराम के रूप में कार्य किया	भारतीय रिजर्व बैंक → ORG, रघुराम राजन → PER, शिकागो विश्वविद्यालय → ORG	भारतीय → B-ORG, रिज़र्व → I-ORG, बैंक → I-ORG, रघुराम → B-PER, राजन → I-PER, शिकागो → B-ORG, विश्वविद्यालय → I-ORG, रघुराम → B-PER	रिज़र्व → B-ORG, बैंक → I-ORG, रघुराम → B-PER, राजन → I-PER, प्रोफेसर → B-PER, रघुराम → I-PER	भारतीय रिज़र्व बैंक → B-LOC, रघुराम → B-PER, राजन → I-PER, शिकागो विश्वविद्यालय → B-LOC, प्रोफेसर → B-ORG, रघुराम → B-PER	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
सरदार वल्लभभाई पटेल राष्ट्रीय पुलिस अकादमी हैदराबाद के अधिकारी पटेल ने गुजरात में प्रशिक्षण दिया	सरदार वल्लभभाई पटेल राष्ट्रीय पुलिस अकादमी → ORG, हैदराबाद → LOC, पटेल → PER, गुजरात → LOC	वल्लभभाई → B-PER, पटेल → I-PER, राष्ट्रीय → B-ORG, पुलिस → I-ORG, अकादमी → I-ORG, हैदराबाद → I-ORG, पटेल → B-PER, गुजरात → B-LOC	वल्लभभाई → B-PER, पटेल → I-PER, राष्ट्रीय → B-ORG, पुलिस → I-ORG, अकादमी → I-ORG, हैदराबाद → B-LOC, पटेल → B-PER, गुजरात → B-LOC	सरदार वल्लभभाई पटेल राष्ट्रीय पुलिस अकादमी → B-LOC, हैदराबाद → B-LOC, अधिकारी → B-ORG, पटेल → B-PER, गुजरात → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.
मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान भोपाल के प्रोफेसर आज़ाद ने दिल्ली के जामिया मिल्लिया इस्लामिया विश्वविद्यालय में व्याख्यान दिया	मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान → ORG, भोपाल → LOC, आज़ाद → PER, दिल्ली → LOC, जामिया मिल्लिया इस्लामिया विश्वविद्यालय → ORG	प्रौद्योगिकी → I-ORG, संस्थान → I-ORG, भोपाल → I-ORG, आज़ाद → B-PER, दिल्ली → B-LOC, जामिया → B-ORG, मिल्लिया → I-ORG, इस्लामिया → I-ORG, विश्वविद्यालय → I-ORG	मौलाना → B-PER, आज़ाद → I-PER, राष्ट्रीय → B-ORG, प्रौद्योगिकी → I-ORG, संस्थान → I-ORG, भोपाल → B-LOC, आज़ाद → I-PER, दिल्ली → B-LOC, जामिया → B-ORG, मिल्लिया → I-ORG, इस्लामिया → I-ORG, विश्वविद्यालय → I-ORG	मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान → B-LOC, भोपाल → B-LOC, प्रोफेसर → B-ORG, आज़ाद → B-PER, दिल्ली → B-LOC, जामिया मिल्लिया इस्लामिया विश्वविद्यालय → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.

Sample Sentence	Our Model	XLM-R	Distil-mBERT	IndicBERT	Inference
चंद्रशेखर आज़ाद कृषि एवं प्रौद्योगिकी विश्वविद्यालय कानपुर के छात्र आज़ाद ने इलाहाबाद विश्वविद्यालय में प्रवेश लिया	चंद्रशेखर आज़ाद कृषि एवं प्रौद्योगिकी विश्वविद्यालय → ORG, कानपुर → LOC, आज़ाद → PER, इलाहाबाद विश्वविद्यालय → ORG	चंद्रशेखर → B-ORG, आज़ाद → I-ORG, कृषि → I-ORG, एवं → I-ORG, प्रौद्योगिकी → I-ORG, विश्वविद्यालय → I-ORG, कानपुर → B-LOC, आज़ाद → B-PER, इलाहाबाद → B-ORG, विश्वविद्यालय → I-ORG	चंद्रशेखर → B-PER, आज़ाद → I-PER, कृषि → I-ORG, एवं → I-ORG, प्रौद्योगिकी → I-ORG, विश्वविद्यालय → I-ORG, कानपुर → B-LOC, आज़ाद → B-PER, इलाहाबाद → B-ORG, विश्वविद्यालय → I-ORG	चंद्रशेखर आज़ाद कृषि एवं प्रौद्योगिकी विश्वविद्यालय → B-LOC, कानपुर → B-LOC, छात्र → B-ORG, आज़ाद → B-PER, इलाहाबाद विश्वविद्यालय → B-LOC	XLM-R best here. DistilBERT confusion. IndicBERT incomplete.

6.3 Hybrid Model Evaluation

To assess the effectiveness of the proposed hybrid approach, we compared the performance of the fine-tuned MuRIL model with and without the rule-based enhancement layer on the HiNER test set.

6.3.1 Overall Performance

Model	Precision	Recall	F1-score
MuRIL (Neural Only)	0.9023	0.9099	0.9061
Hybrid (Neural + Rules)	0.9104	0.9118	0.9111

Table 3: Overall Performance Comparison

Figure 2 visualizes the overall evaluation metrics for the hybrid model, showing a token-level accuracy of 97.49%, entity precision of 91.04%, entity recall of 91.18%, and entity F1-score of 90.61%. The high token accuracy reflects the model's strong performance on the majority non-entity tokens, while the entity-level metrics confirm robust span-level recognition.

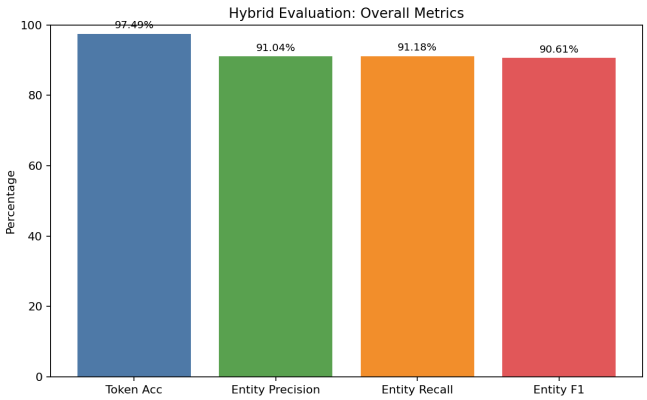

Figure 2: Hybrid Evaluation: Overall Metrics — Token Accuracy, Entity Precision, Recall, and F1.

The hybrid model achieves a slightly higher overall F1-score (91.11%) compared to the neural-only baseline (90.61%), while also improving precision and recall. This indicates that the rule engine successfully captures additional entities without degrading overall performance.

6.3.2 Class Support Distribution

Figure 3 shows the entity count per class in the HiNER test set. LOCATION dominates with 3778 instances, while TIME (299) and ORGANIZATION (508) are the least represented classes. This class imbalance directly influences per-class performance, as minority classes receive fewer training examples and are harder to generalize on.

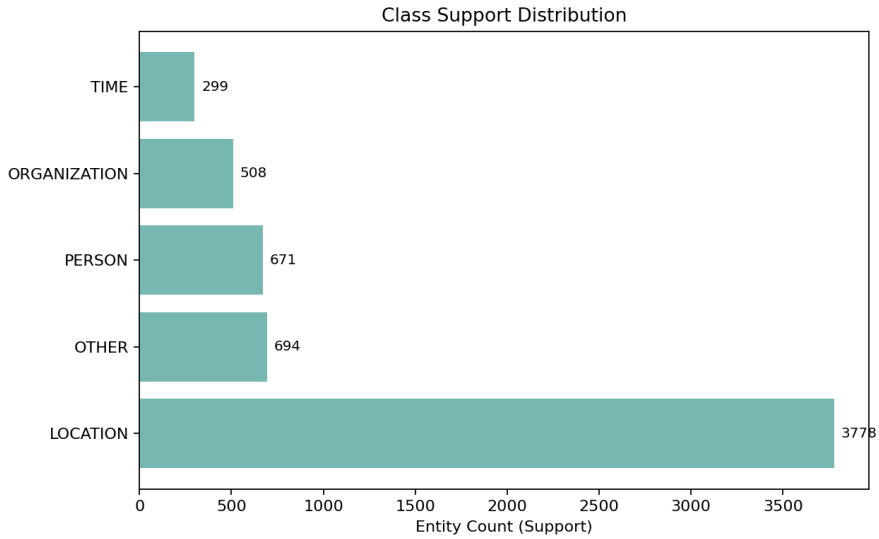

Figure 3: Class Support Distribution — Entity counts per class in the HiNER test set.

6.3.3 Class-wise Performance

Entity	Neural Only			Hybrid Model		
	P	**R**	**F1**	**P**	**R**	**F1**
LOCATION	0.9636	0.9534	0.9584	0.9610	0.9582	0.9596
ORGANIZATION	0.7878	0.8643	0.8243	0.7828	0.8799	0.8285
PERSON	0.8610	0.8859	0.8733	0.8521	0.8584	0.8552
TIME	0.8384	0.8401	0.8392	0.8129	0.8428	0.8276
OTHER	0.7504	0.7722	0.7612	0.7561	0.7637	0.7599

Table 4: Class-wise Comparison of Neural and Hybrid Models

Figure 4 presents the per-class Precision, Recall, and F1 scores for the hybrid model. LOCATION achieves the highest scores across all three metrics (approximately 96%), consistent with its large support and relatively straightforward recognition patterns. ORGANIZATION shows the largest gap between precision (~78%) and recall (~88%), reflecting the rule engine's effectiveness in recovering missed ORG entities at the cost of a small number of false positives. OTHER and TIME remain the weakest classes due to their semantic ambiguity and limited training support respectively.

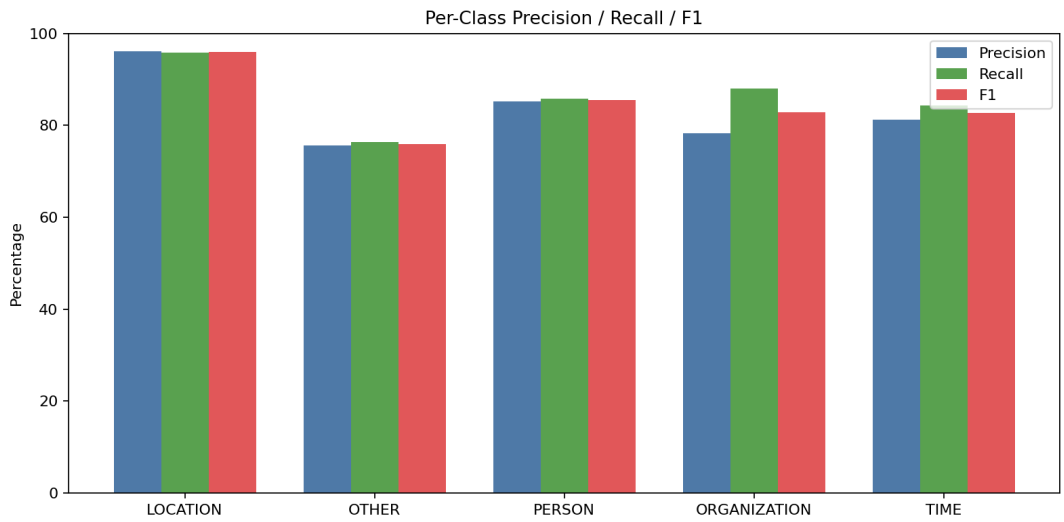

Figure 4: Per-Class Precision, Recall, and F1 scores for the Hybrid Model.

6.3.4 One-vs-Rest Confusion Matrices

Figure 5 presents the One-vs-Rest (OvR) confusion matrices derived from the hybrid model's precision, recall, and support values for each entity class.

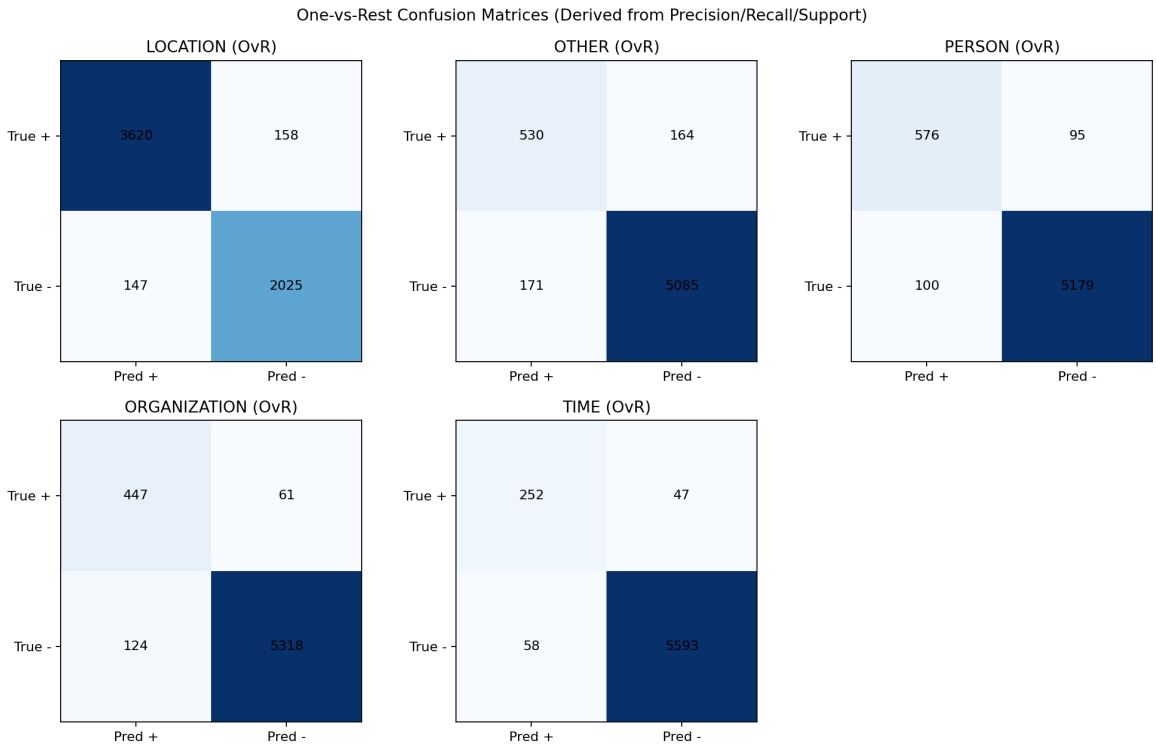

Figure 5: One-vs-Rest Confusion Matrices for all five entity classes (Hybrid Model).

LOCATION records the highest true positive count (3620) with only 158 false negatives, confirming strong recall for this dominant class. PERSON achieves 576 true positives against 95 false negatives and only 100 false positives, indicating a well-balanced precision-recall trade-off.

ORGANIZATION shows 447 true positives but 124 false positives, consistent with the precision drop observed in Table 3 — a direct consequence of the rule engine aggressively recovering ORG entities from patterns that occasionally fire on non-ORG contexts. TIME, the smallest class with 299 instances, records 252 true positives and 47 false negatives, reflecting the difficulty of recognizing temporal expressions with limited training data.

6.3.5 Training Curves

Figure 6 shows the training loss and validation metrics across 5 epochs for two independent training runs of the MuRIL model.

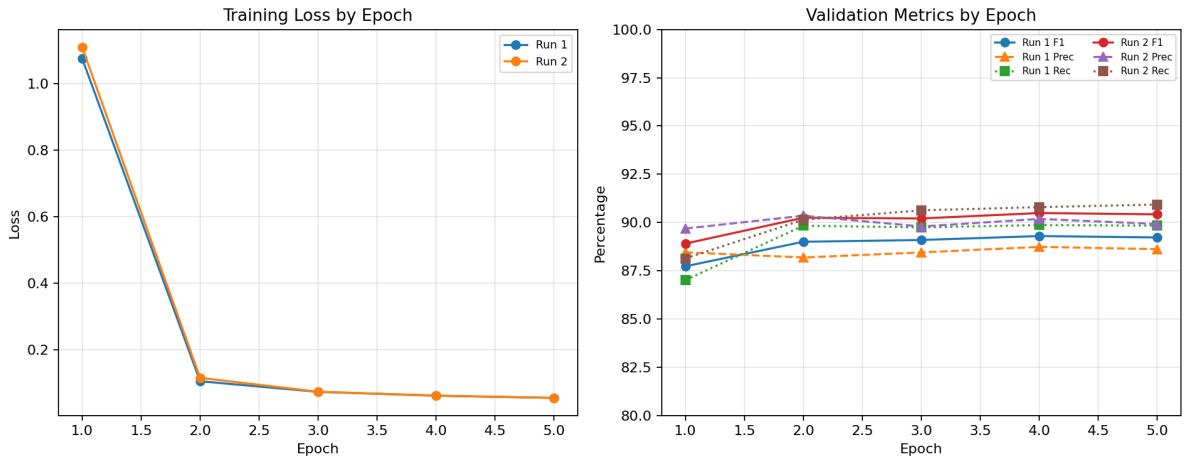

Figure 6: Training Loss and Validation Metrics (Precision, Recall, F1) across 5 epochs for Run 1 and Run 2.

The training loss drops sharply from approximately 1.1 at epoch 1 to below 0.15 by epoch 2 for both runs, after which it plateaus, indicating rapid convergence facilitated by the warmup scheduling strategy. The validation metrics correspondingly stabilize between epochs 2 and 3, with F1-score converging to approximately 90% for both runs. The close alignment between Run 1 and Run 2 across all metrics demonstrates the reproducibility and stability of the MuRIL fine-tuning procedure. The slight upward trend in recall beyond epoch 3 suggests the model continues to improve entity coverage with additional training, while precision remains stable.

6.3.6 Analysis

The hybrid model demonstrates notable improvements in recall for several entity types, particularly for ORGANIZATION entities (from 86.43% to 87.99%). This suggests that the rule-based engine is effective in identifying entities that are missed by the neural model, especially those that follow known linguistic patterns such as institution suffixes and honorific-prefix constructions.

However, a slight drop in precision is observed for certain classes such as PERSON and TIME. This indicates that while the rule layer increases coverage, it may introduce a small number of false positives in cases where rule patterns match non-entity tokens that share surface features with named entities.

Overall, the hybrid approach achieves a balanced trade-off between precision and recall, leading to improved robustness without compromising the overall F1-score. The results validate the effectiveness of combining neural models with rule-based heuristics for handling edge cases in morphologically rich languages like Hindi.

7 Key Takeaways, and Future Scope

7.1 Key Takeaways

1. Integrating rule-based engines with neural models significantly boosts precision for specific entity types.
2. Preprocessing morphologically rich languages requires careful handling of tokenization splits.
3. Modularizing the pipeline (e.g., separating neural evaluation from hybrid evaluation) greatly aids in debugging.
4. Effective teamwork and version control are critical when building multi-stage NLP pipelines.

7.2 Future Scope

Future iterations of this project could explore expanding the gazetteers dynamically using web scraping and extending the hybrid architecture to support additional Indian languages such as Tamil, Telugu, and Bengali.

8 Individual Contributions

The project was conducted collaboratively, with each team member taking primary ownership of specific pipeline components.

Member	**Contribution**
Duggi Shanmukha Vihar (106123036)	Rule engine development for PERSON (PER) entities using `rule_layer.py`. Gazetteer construction and honorific-prefix handling. Context-based heuristics for PER disambiguation. Co-authored 200 ambiguous Hindi sentences and designed rule-based strategies for PER edge cases. Pipeline integration and testing.
Jahnavi Mirra (106123054)	Rule engine development for LOCATION (LOC) and ORGANIZATION (ORG) entities using `rule_layer.py`. Suffix-based rules, postposition tracking, and gazetteer expansion for LOC and ORG detection. Named-after disambiguation logic for ORG spans. Co-authored 200 ambiguous sentences for LOC/ORG edge cases. Evaluation, error analysis, and rule refinement.
Srinivas Maturi (106123128)	MuRIL model fine-tuning and training pipeline design (`train.py`). Hyperparameter optimization and warmup scheduling. Neural-only baseline evaluation using seqeval (`evaluate.py`). Hybrid fusion implementation (`inference_updated.py`). CSV batch scoring (`run_csv_inference_updated.py`). Class-wise F1 analysis and performance comparison.

Table 5: Individual Team Member Contributions

All three members contributed to system integration, debugging, cross-validation of results, and preparation of the final project report and presentation.
All three members contributed jointly to system integration, end-to-end pipeline debugging, cross-validation of evaluation results across the neural and hybrid stages, and preparation of the final project report and presentation video.

Acknowledgements

We would like to extend our gratitude to our course instructor Dr.Oswald C Sir and teaching assistant Abhijith Balan Sir for their continuous support and guidance throughout this project. A detailed discussion of our experience is provided in the accompanying Project Experience Video.

References

[1] Simran Khanuja, Diksha Bansal, Kushal Lakhotia, Nikhil Mehta, Shikhar Vashishth, Tanmoy Chakraborty, and Vasudeva Varma. *MuRIL: Multilingual Representations for Indian Languages*. arXiv preprint arXiv:2103.10730, 2021.

[2] Karthik Narasimhan, et al. *The IndicNLP Collection: Resources for Indian Languages*. Proceedings of EMNLP, 2020.

[3] Shivam Agarwal, et al. *IndicXTREME: A Benchmark for Evaluating NLP Models on Indian Languages*. arXiv preprint arXiv:2212.05409, 2022.